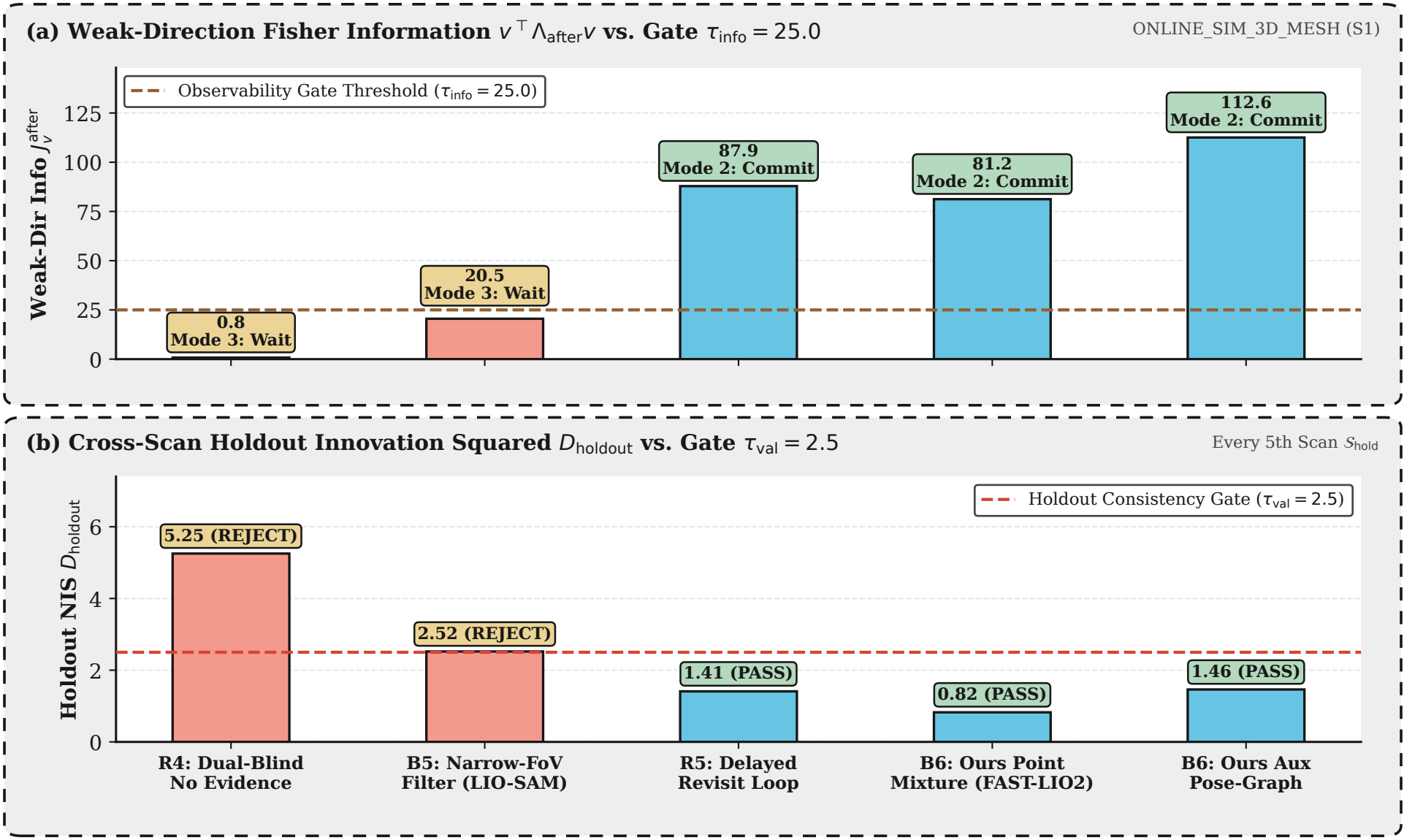
(a) Weak-Direction Fisher Information $v^\top \Lambda_{\text{after}} v$ vs. Gate $\tau_{\text{info}} = 25.0$
ONLINE_SIM_3D_MESH (S1)
Observability Gate Threshold ($\tau_{\text{info}} = 25.0$)
Weak-Dir Info J_v^{after}
125
100
75
50
25
0
0.8
Mode 3: Wait
20.5
Mode 3: Wait
87.9
Mode 2: Commit
81.2
Mode 2: Commit
112.6
Mode 2: Commit
(b) Cross-Scan Holdout Innovation Squared D_{holdout} vs. Gate $\tau_{\text{val}} = 2.5$
Every 5th Scan $\mathcal{S}_{\text{hold}}$
Holdout Consistency Gate ($\tau_{\text{val}} = 2.5$)
Holdout NIS D_{holdout}
6
4
2
0
5.25 (REJECT)
2.52 (REJECT)
1.41 (PASS)
0.82 (PASS)
1.46 (PASS)
R4: Dual-Blind No Evidence
B5: Narrow-FoV Filter (LIO-SAM)
R5: Delayed Revisit Loop
B6: Ours Point Mixture (FAST-LIO2)
B6: Ours Aux Pose-Graph